Customer Churn Analysis (No ML – Pure Data Analysis)

🎯 Project Goal

Identify patterns between:

- Customers who **churned (left)**
- Customers who **stayed**

No machine learning. Only:

- Filtering
- Grouping
- Aggregation
- Comparison
- Visualization
- Business insights

🧩 STEP 1: Understand the Dataset

Typically churn datasets contain columns like:

Column	Meaning
customerID	Unique ID
gender	Male/Female
SeniorCitizen	0/1
tenure	Months with company
MonthlyCharges	Monthly bill
TotalCharges	Total paid
Contract	Month-to-month / One year / Two year
PaymentMethod	Credit Card / Electronic Check etc
InternetService	DSL / Fiber / No
Churn	Yes / No

🛠 STEP 2: Load & Inspect Data (Python)

```
import pandas as pd

df = pd.read_csv("customer_churn.csv")

df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   customerID        7043 non-null   str
 1   gender            7043 non-null   str
 2   SeniorCitizen     7043 non-null   int64
 3   Partner           7043 non-null   str
 4   Dependents        7043 non-null   str
 ..  ........          ...  ....       ..
17   PaymentMethod     7043 non-null   str
18   MonthlyCharges    7043 non-null   float64
19   TotalCharges      7043 non-null   str
20   Churn             7043 non-null   str
dtypes: float64(1), int64(2), str(18)
memory usage: 1.1 MB
```

Check:

- Missing values
- Data types
- Incorrect formatting (TotalCharges often string)

🧹 STEP 3: Data Cleaning

1️⃣ Fix TotalCharges

```
df['TotalCharges'] = pd.to_numeric(df['TotalCharges'], errors='coerce')
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   customerID        7043 non-null   str
 1   gender            7043 non-null   str
 ..  .........         ...  ....       ..
 9   TotalCharges      7032 non-null   float64
20   Churn             7043 non-null   str
dtypes: float64(1), int64(2), str(18)
memory usage: 1.1 MB
```

```
df.isnull().sum()
```

```
customerID        0
gender            0
SeniorCitizen     0
.................  ...
MonthlyCharges    0
TotalCharges     11
Churn             0
dtype: int64
```

2️⃣ Handle missing rows

```
df.dropna(inplace=True)
df.isnull().sum()
```

```
customerID        0
gender            0
SeniorCitizen     0
.................  ...
MonthlyCharges    0
TotalCharges      0
Churn             0
dtype: int64
```

Why STEP 3 (Data Cleaning) is Important?

Real datasets are **never perfectly clean**.

In the IBM Telco dataset:

- TotalCharges looks numeric
- But actually stored as **object (string)**
- Some rows contain **blank spaces**

That causes:

- Calculation errors
- Wrong averages
- Model failures later

🧩 PART 1 – Fix TotalCharges

Code:

```
df['TotalCharges'] = pd.to_numeric(df['TotalCharges'], errors='coerce')
```

🔍 What is happening here?

Problem: TotalCharges as **object/str**

It should be **float**, not object.

Why object?
Because some rows contain: `" "` (blank space)

So **pandas treats** entire column as **string**.

🧠 What pd.to_numeric() Does

It converts column into numbers.

Parameter: `errors='coerce'`

Means: `If conversion fails → replace with NaN`

So:

Original Value	After Conversion
"100.25"	100.25
"2500.00"	2500.00
" " (blank)	NaN

So bad values become NaN.

🧩 PART 2 – Drop Missing Rows

df.dropna(inplace=True)

🔍 What does this do?

It removes rows where any column has NaN.

So those 11 bad rows get deleted.

🔎 What is inplace=True?

It means: `Modify original dataframe directly.`

Without it: `df = df.dropna()`

Both are correct.

🔥 Better Analytical Approach (Pro Version)

Instead of blindly dropping:

Check first: `df[df['TotalCharges'].isnull()]`

You'll notice:
- tenure = 0
- churn = Yes mostly

So dropping them is safe because:
- Very small number
- Doesn't affect analysis much

⚠️ Important Interview Question

Interviewer may ask:

Q: Why did you use errors='coerce'?
Answer:
Because TotalCharges column had some non-numeric blank values. Using errors='coerce' safely converts invalid entries into NaN, which allows us to identify and handle missing values systematically.

Q: Why are TotalCharges missing?
Answer:
Usually customers with:
- tenure = 0 (tenure = Number of months the customer has stayed with the company)

tenure	Meaning
1	Customer joined 1 month ago
12	1 year old customer
60	5 years old customer
0	Just joined (0 months completed)

- Just signed up
- No billing yet

So TotalCharges = MonthlyCharges × tenure
If tenure = 0 → TotalCharges = blank

➔ Why Would a Customer Churn at tenure = 0?

Good analytical thinking here 👇

Possibilities:
1. Signed up but canceled immediately
2. Service activation failed
3. Customer changed mind
4. Data entry issue

In telecom, sometimes:
- Customer registers
- Installation not done
- Account closed

So churn = Yes but tenure = 0

🚀 For You (Advanced Practice)

In SQL this would be like: **`SELECT * FROM customers WHERE TotalCharges IS NULL;`**

Same thinking — different tool.

🔍 **STEP 4: Basic Churn Distribution**

🔷 **How many customers churned?**

```
df['Churn'].value_counts()
```

```
Churn
No    5163
Yes   1869
Name: count, dtype: int64
```

✅ **What this means:**
Out of all customers:

- **5163 customers stayed**
- **1869 customers left (churned)**

So total customers: 5163 + 1869 = ***7032 customers***

```
df['Churn'].value_counts(normalize=True) * 100
```

```
Churn
No    73.421502
Yes   26.578498
Name: proportion, dtype: float64
```

✅ **What this means:**

- **73.42% customers stayed**
- **26.58% customers churned**

So roughly: 🔴 ***1 out of every 4 customers leaves the company***

📊 **Business Interpretation (Very Important)**

Now think like a Business Analyst: **Is 26.58% churn high?**
In telecom industry:

- 10–15% → Norma
- 20% → Concerning
- 25%+ → High churn problem

So:

- 👉 26.58% is high
- 👉 Company is losing more than 1/4 customers

NOTE:- *That's a serious revenue issue.*

💰 **Business Impact Example**

- Imagine: **7000 customers**
  - Average MonthlyCharges = ₹70
  - 1869 customers leave
  - Monthly revenue lost: **1869 × 70 = ₹130,830 per month**
  - Annual loss: **₹130,830 × 12 = ₹15,69,960**

Now you see why churn analysis matters 🔥

📌 **Insight:**

- What % of customers leave?
- Is churn high (>25%)?

📊 STEP 5: Feature Comparison (Core Part)

Now we compare churn vs non-churn customers.

5.1 📈 Churn by Contract Type

```
pd.crosstab(df['Contract'], df['Churn'], normalize='index') * 100
```

What This Code Does

- pd.crosstab() → creates a comparison table
- Rows = Contract type
- Columns = Churn (Yes/No)
- normalize='index' → calculate percentage within each contract type
- 100 → convert to %

📊 Typical Output (Example)

Contract	No (%)	Yes (%)
Month-to-month	57%	43%
One year	89%	11%
Two year	97%	3%

🔎 Interpretation

- 👉 Month-to-month customers churn the most
- 👉 2-year contract customers rarely churn

💼 Business Insight

Long-term contracts create **customer lock-in & loyalty**.

Conclusion: Customers without long-term commitment are high churn risk.

📌 Insight Questions:

- Is churn highest in Month-to-month?
- Are long contracts safer?

5.2 💰 Churn by Monthly Charges

```
df.groupby('Churn')['MonthlyCharges'].mean()
```

What This Code Does

- Groups customers by churn (Yes/No)
- Calculates average monthly charge for each group

📊 Example Output

Churn	Avg Monthly Charges
No	61
Yes	74

🔎 Interpretation

Customers who churn pay higher bills.

💼 Business Insight

High monthly charges may cause dissatisfaction.

Possible reasons:

- Expensive fiber plans
- Add-on services
- Price sensitivity

Conclusion:

Price sensitivity is a churn driver.

📌 **Ask:**

- Do high-paying customers churn more?

5.3 ⏳ Churn by Tenure

```
df.groupby('Churn')['tenure'].mean()
```

Example Output

Churn	Avg Tenure
No	37 months
Yes	18 months

🔎 **Interpretation**

Churned customers stayed for fewer months.

💼 **Business Insight**

New customers are risky.

Common pattern in telecom:

📌 First 6–12 months = highest churn risk

After 2+ years:

- Customers are stable
- Switching cost increases

Conclusion:

Early lifecycle engagement is critical.

📌 **Ask:**

- Do new customers churn more?
- Are loyal customers stable?

5.4 👴 Senior Citizen Impact

```
pd.crosstab(df['SeniorCitizen'], df['Churn'], normalize='index') * 100
```

What It Does

Compare churn % between:

- Senior citizens (1)
- Non-seniors (0)

Example Output

SeniorCitizen	No	Yes
0	75%	25%
1	58%	42%

Interpretation

Senior citizens churn more.

Business Insight

Possible reasons:

- Pricing issues
- Technology discomfort
- Service complexity

5.5 💳 Payment Method Impact

```
pd.crosstab(df['PaymentMethod'], df['Churn'], normalize='index') * 100
```

Electronic check usually shows high churn 👀

Example Output

Payment Method	Churn %
Electronic check	45%
Mailed check	20%
Credit card (auto)	15%
Bank transfer (auto)	18%

🔎 **Interpretation**

Electronic check customers churn the most.

💼 **Business Insight**

Electronic check users:

- Not on auto-payment
- Less committed
- Possibly dissatisfied

Auto-payment users:

- Lower churn
- Higher retention

Strong Business Conclusion:

Encouraging auto-pay reduces churn.

Interview-Level Answer

Q: What were key churn drivers?

You can say:

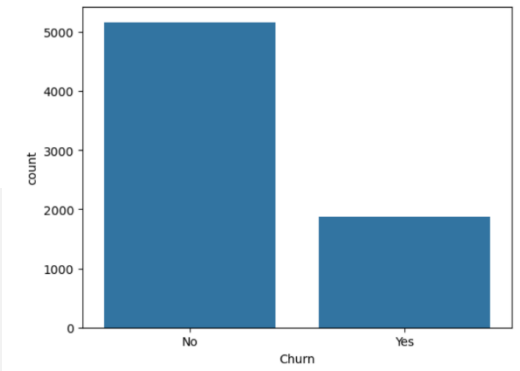
Customers on month-to-month contracts, paying higher monthly charges, with lower tenure, and using electronic check payment methods showed significantly higher churn rates.

📊 STEP 6: Visual Analysis

Now we visualize patterns.

1 Churn Count Plot

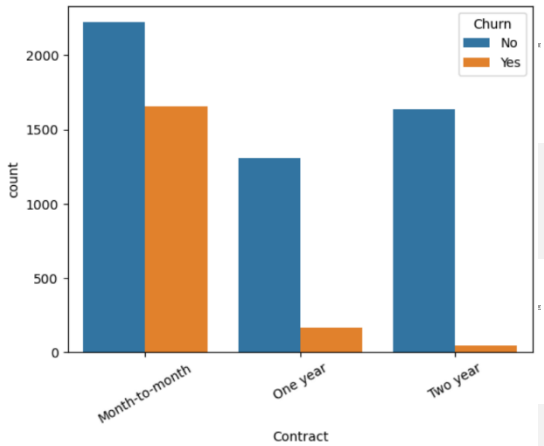
```
import seaborn as sns
import matplotlib.pyplot as plt

sns.countplot(x='Churn', data=df)
plt.show()
```



---

2 Contract vs Churn

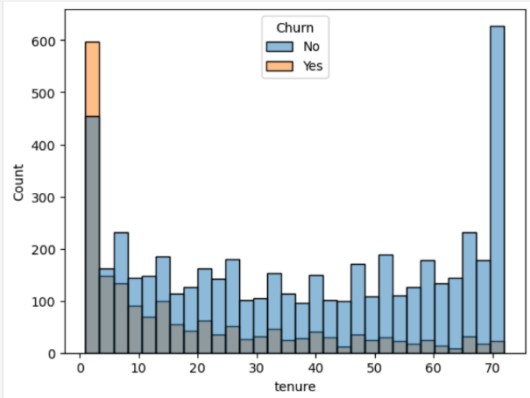
```
sns.countplot(x='Contract', hue='Churn', data=df)
plt.xticks(rotation=30)
plt.show()
```



---

3 Tenure Distribution

```
sns.histplot(data=df, x='tenure', hue='Churn', bins=30)
plt.show()
```



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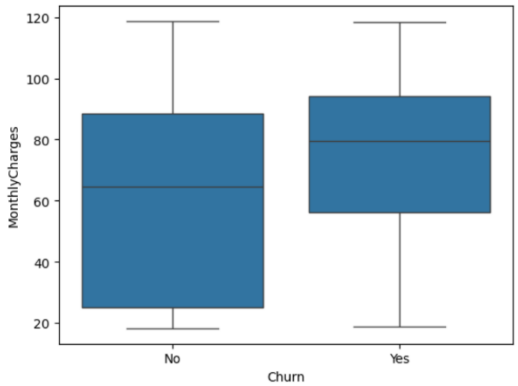
4 Monthly Charges Boxplot

```
sns.boxplot(x='Churn', y='MonthlyCharges', data=df)
plt.show()
```



🧠 STEP 7: Business Insights (Most Important Part)

Now think like a Business Analyst, not a coder.

Example insights:

Insight 1: Month-to-month customers churn 3x more than 2-year contract customers.

Insight 2: Customers with high monthly charges churn more frequently.

Insight 3: Customers with tenure < 12 months are high-risk.

Insight 4: Electronic check users churn more.

📈 STEP 8: Risk Segmentation (Advanced Analysis – No ML)

Create simple churn risk logic.

Example:

```
df['RiskLevel'] = 'Low'

df.loc[(df['Contract']=='Month-to-month') & (df['tenure'] < 12) & (df['MonthlyCharges'] > 70), 'RiskLevel'] = 'High'
```

Now check:

```
pd.crosstab(df['RiskLevel'], df['Churn'], normalize='index') * 100
```

Output:

Churn RiskLevel	No	Yes
High	30.466830	69.533170
Low	79.044709	20.955291

🔥 This is **business rule-based churn prediction** (very strong in interviews).

Notes:-

1. What is Risk Segmentation?

Risk segmentation means:

Dividing customers into **risk groups** based on their likelihood of leaving.

Example:

Customer	Risk Level
Long contract + low price	Low risk
Month-to-month + expensive	High risk

Companies use this to **target retention campaigns**.

2. Define High Risk Customers

Code:

```
df.loc[(df['Contract']=='Month-to-month') & (df['tenure'] < 12) & (df['MonthlyCharges'] > 70), 'RiskLevel'] = 'High'
```

Meaning

We are saying:

If a customer satisfies **all these conditions**, mark them as **High risk**.

Condition:-

Condition	Reason
Month-to-month contract	Easy to leave
tenure < 12 months	New customers leave more
MonthlyCharges > 70	Expensive plans cause churn

Example:-

Contract	tenure	MonthlyCharges	RiskLevel
Month-to-month	5	85	High
Month-to-month	8	75	High
Two year	30	60	Low
One year	18	65	Low

So you **segmented customers based on churn risk**.

3. Now Check if Your Logic Works

```
pd.crosstab(df['RiskLevel'], df['Churn'], normalize='index') * 100
```

This compares:

- RiskLevel
- Actual churn

4. Your Output

RiskLevel	No	Yes
High	30.46%	**69.53%**
Low	**79.04%**	20.95%

5. Interpretation

a. High Risk Customers

Result: 69.5% actually churn

- That means **your rule works very well**.
- Most customers identified as **High risk actually left**.

b. Low Risk Customers

Result: Only 20.9% churn

- Meaning these customers are **more stable**.

6. Business Meaning

Your segmentation successfully identifies:

Segment	Churn Probability
High Risk	~70%
Low Risk	~21%

So the company can focus retention on **high-risk customers**.

Example actions:

- Discount offers
- Contract upgrade
- Loyalty programs

7. Why This Is Powerful (Even Without ML)

This is called: **Rule-based predictive analytics**

Companies do this when:

- No ML model exists yet
- Business wants quick insights

It is **very common in telecom, banking, and e-commerce**.

📝 STEP 9: Final Project Structure (For Resume)

When you write this in resume:

> Performed exploratory churn analysis on telecom customer dataset using Pandas and Seaborn. Identified key churn drivers including contract type, tenure, and payment method. Designed rule-based churn risk segmentation improving business interpretability.

Churn VS **Attrition**

Churn and **Attrition** are very similar concepts but used in **different contexts**. Both mean **losing something over time**, but the subject differs.

1️⃣ Customer Churn

Used in:

- Telecom
- Banking
- SaaS
- Subscription services

Definition: Customer churn = when customers stop using a company's product or service.

Example:

- A telecom customer cancels their mobile plan
- Netflix user cancels subscription
- Bank customer closes account

Example Table

Customer	Status
Customer A	Active
Customer B	Cancelled → **Churn**
Customer C	Active

So:

Churn = Losing customers

2️⃣ Employee Attrition

Used in:

- HR analytics
- Workforce management

Definition

Attrition = when employees leave an organization.

Example:

- Employee resigns
- Employee retires
- Employee quits job

Example Table

Employee	Status
Employee A	Working
Employee B	Resigned → **Attrition**
Employee C	Working

So:

Attrition = Losing employees

---

3 Key Difference

Feature	Churn	Attrition
Used in	Customer analytics	HR analytics
Refers to	Customers leaving	Employees leaving
Example	Cancel subscription	Resign job

---

4 In Data Analysis

Customer churn analysis

Focus on:

- Contracts
- Pricing
- Customer behavior
- Service quality

Example metric:

Churn Rate = Customers lost / Total customers

Employee attrition analysis

Focus on:

- Salary
- Work environment
- Job satisfaction
- Work-life balance

Example metric:

Attrition Rate = Employees left / Total employees

---

5 Example Interview Question

Q: What is the difference between churn and attrition?

Answer:

Churn refers to customers discontinuing a service or subscription, while attrition refers to employees leaving an organization. Both measure loss over time but apply to different domains.